A DISTINCTION BETWEEN BED-LOAD AND SUSPENDED LOAD IN NATURAL STREAMS

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Introduction--Problems pertaining to the transportation of sediment by flowing water have been extensively studied by engineers and scientists during the last few years. Most of the various experiments have been concerned with either bed-load or suspended load, but seldom have they been studied simultaneously.

When bed-load and suspended load occur simultaneously in a flume or stream a distinct difference in their mechanical composition is often observed. A curve fitted to the plotted data from a mechanical analysis of the bed-material shows a distinct break. This break occurs most commonly in the data plotted from samples taken in mountain streams where the bed-material is extremely coarse. For instance, in the Rhine River, Switzerland [see 5 of "References" at end of paper] the bed-material ranges in size from pebbles 5 mm in diameter to boulders 100 mm in diameter and greater. Between boulders and in protected pools, however, sand one mm in diameter and finer is found, whereas particles ranging in size from one to five mm are not present. The explanation for this size-distribution has been that the material greater than five mm in diameter moves as bed-load and the material finer than one mm in diameter moves as suspended load. Particles ranging in size from one mm to five mm in diameter occur rarely for certain geologic or hydraulic reasons [3].

Observations indicate that the bed-load of streams is pushed, rolled, or saltated along the bed, while the suspended load is distributed throughout the whole cross-section. This difference in method of transportation has been made a criterion for the distinction between bed-load and suspended load; that is, bed-load has been defined as the relatively coarse part of the sediment-load that moves along the stream-bed, and suspended load as the fine part that is distributed throughout the cross-section.

During the past 30 years, many formulas for the rate of transportation in terms of various flow- and sediment-characteristics have been introduced for bed-load, but little has been done in this respect as to suspended load. On the other hand, research has proposed laws for the distribution of suspended load over a cross-section, but no similar expression has been developed for bed-load. The difference in behavior of the different loads is the main reason for a distinction being made between bed-load and suspended load, even though the definition of the distinction often leads to difficulties. Inconsistencies occur when the bed consists of fine sand, especially when the suspended load is of the same or of a similar mechanical composition as the bed-load, and there is no marked boundary between the two.

Although the same material moves partly in suspension, it is still convenient to make a distinction between the two modes of transportation, because the two loads are measured by different methods. Suspended matter, with few exceptions, is measured in terms of concentration by taking samples of the water carrying the sediment. Bed-load, however, cannot be measured in terms of concentration because close to the bed the movement of sediment and water is uncertain. Instead, the rate of bed-load transportation is determined directly by a suitable trap that permits water to pass through but retains the solids by screens or by locally reducing the water-velocity inside the trap. Obviously the distinction between suspended load and bed-load in most sediment-load data is, in reality, merely a distinction in methods of experimental observation. This distinction was employed by Shaank and Slotboom [6] in their studies of the lower Rhine.

As pointed out, the usual distinction between bed-load and suspended load has serious limitations. Observations in certain streams indicate that it is of even less value when the relatively coarse part of the suspended load moves at rates that can be expressed in terms of the stream-discharge, while the relatively fine part bears no relation whatsoever to discharge. For instance, Vetter [7] found that in the Colorado River all the sizes in suspension that show a relation with discharge are also freely available in the bed.

Soil Conservation Service sediment-load investigations--In formulating a program for the Soil Conservation Service, it was recognized that there was urgent need for extensive information on the factors affecting the transportation of sediment in natural streams. A comprehensive program of research therefore was planned, resulting in the construction of a sediment-load laboratory on the Enoree River near Greenville, South Carolina [2]. The most important feature of the laboratory is a new type of control which permits the measurement of the total sediment-load of the river. As in previous studies made on the measurement of the total sediment-load of streams [6], all sediment in transportation could not be measured by any one sampling method.

Two sampling procedures are used in the operation of the control to determine the total sediment-load. One portion of the load, consisting of the relatively coarse fractions moving on or near the bed, is removed hydraulically by pipe-connection and pump from openings in the stream-bed. This material is discharged into a settling tank, from which it is later removed to be weighed and analyzed for particle-size. The remaining portion of the stream-load, consisting of the material that does not pass into the bottom-openings, is determined by suspended-load samples taken with special multiple-unit samplers [for description of sampler see reference 2]. This material is also subjected to an analysis for the determination of particle-size.

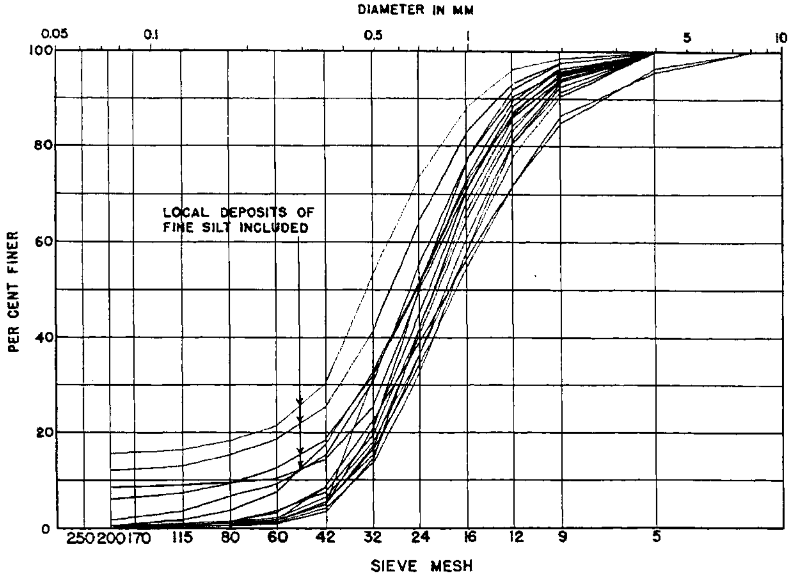

FIG. 1 - AVERAGE COMPOSITION OF BED-MATERIAL ON VARIOUS RANGES ON THE ENOREE RIVER

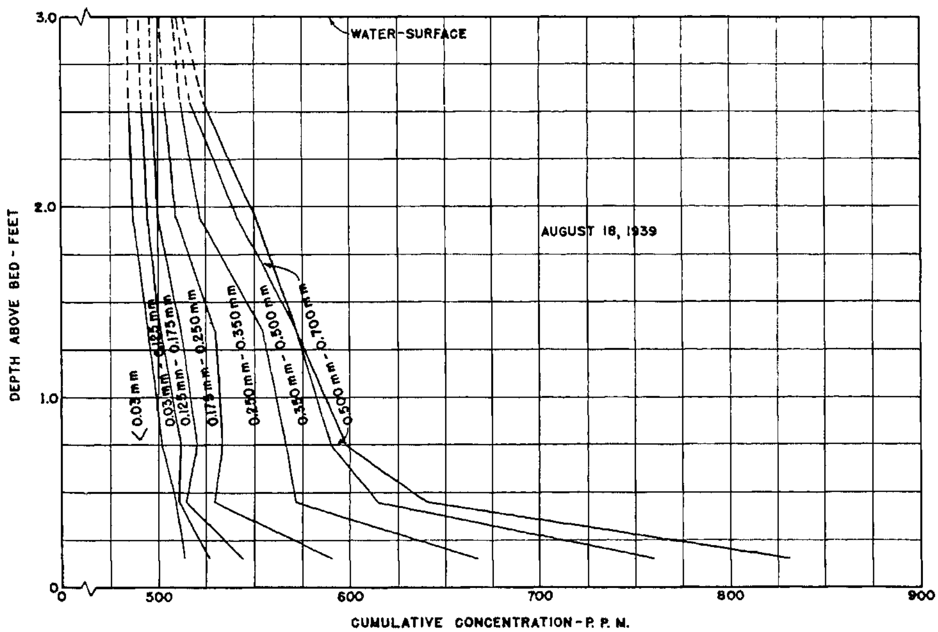

FIG. 2 - TYPICAL SEDIMENT-DISTRIBUTION CURVES IN ENOREE RIVER, S.C.

Several times a year a systematic bed-sampling program is conducted for the purpose of evaluating changes in bed-composition. The procedure of sampling is to scoop a pint container full of material from the bed-surface at four-foot intervals on each of 17 ranges approximately evenly spaced over a 1-1/2-mile reach upstream from the control. The average width of the stream in this reach is 40 feet. Each sample is analyzed for particle-size, and an average mechanical-analysis curve for each range is constructed.

Figure 1, which shows the average mechanical-analysis curves for the various ranges, is typical of the results of a sampling series. Examination of these curves shows that very little material finer than a 42-mesh sieve size (0.351 mm) is available in the bed. A few of the curves, as indicated in Figure 1, show the presence of a considerable quantity of fine material. This fine material was found in samples taken close to the banks and is believed to be the result of local bank undercutting and caving. On the average, however, a break occurs in each curve at

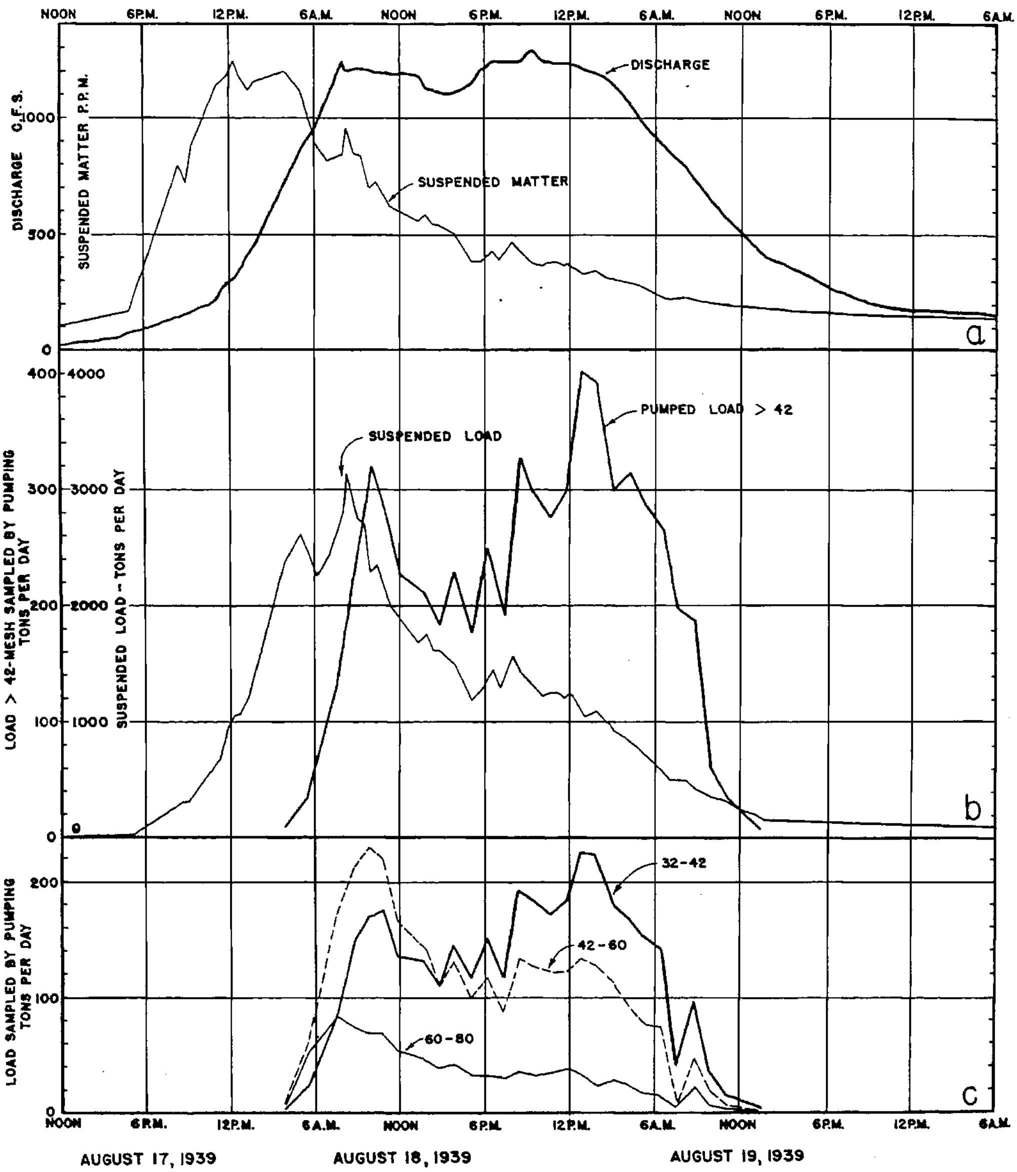

FIG. 3 - DISCHARGE AND SOLIDS-LOAD IN ENOREE RIVER DURING FLOOD OF AUGUST 17, 18, 19, 1939.

the 42-mesh sieve size and indicates that only material coarser than this size is available in the bed in sufficient quantity to be considered abundant. Practically no material finer than the 60-mesh sieve size is found in the bed.

In addition to measurement of sediment-load in the operation of the control, determinations are made of discharge, water-surface slope, dissolved load, rainfall, and water- and air-temperatures. For the purpose of observing changes in the position of the stream-bed, periodic cross-sections are taken on 62 ranges, evenly spaced, over the 1-1/2-mile reach in which samples of bed-material are taken.

Measurements taken during the flood of August 17-19, 1939--A flood of 36-hour duration occurred after a general rainfall of 2.81 inches on August 17 and 1.62 inches on August 18 over the 64.4-square-mile drainage-basin above the control. During this period, sampling of the sediment-load was conducted by the two prescribed methods. A continuous record of discharge was obtained by use of an automatic stage-recorder.

The samples of sediment obtained by pumping were analyzed and the rate of transportation of each size was computed. As suggested by the results of the bed-sample curves (Fig. 1), the rates were summarized for the following typical size fractions: All material greater than 42-mesh sieve; all material passing 32-mesh sieve and retained on 42-mesh sieve; all material passing 42-mesh sieve and retained on 60-mesh sieve; all material passing 60-mesh sieve and retained on 80-mesh sieve.

Frequent samples were taken to determine the average concentration of sediment in suspension throughout the depth of the stream. A typical vertical distribution of various particle-sizes in suspension is shown in Figure 2.

Figure 3a shows a plot of the stream-discharge and the concentration in parts per million of material in suspension. Figure 3b shows a plot of the suspended load and the material coarser than a 42-mesh sieve that is obtained by pumping, both being expressed as a rate in tons per day. The former curve is calculated directly from the curves of Figure 3a. Figure 3c shows the rate of transportation of the various size-fractions mentioned above.

The important features of the various figures may be summarized as follows:

(1) Although there is a general increase in silt-concentration in the river with an increase in water-discharge, the peak silt-concentration precedes the peak water-discharge (Fig. 3a). In the Enoree River the peaks occur in this sequence; although it has been observed in some rivers that occasionally the peak silt-discharge precedes and at other times follows the peak water-discharge, depending largely upon the part of the drainage-area contributing the water and the relative velocity of the water and wave-crest [4].

(2) The rate of transportation of material coarser than a 42-mesh sieve follows a distinctly different curve than that for suspended load (Fig. 3b). (Mechanical analyses showed that the suspended load consisted of material of which 96 per cent is finer than the 42-mesh sieve size.)

(3) With reference to Figure 3c it has been observed that the rate of transportation of material of the sizes ranging between the 32- and 42-mesh sieve follows a curve very similar to the curve (Fig. 3b) for the material coarser than the 42-mesh sieve. On the other hand, material of the size between the 60- and 80-mesh sieve is similar to the curve for suspended load (Fig. 3b). Material of the size between the 42- and 60-mesh sieve size is a transition between the two, but is apparently more similar to the suspended-load curve.

(4) As seen in Figure 1 the average mechanical-analysis curves of the bed-samples from all ranges, except for those samples that include local silt-deposits, show a distinct break at the 42-mesh sieve size and include not more than seven per cent of the material finer than that size.

(5) Particles much coarser than the 42-mesh sieve size are found in suspension, with the coarser fractions more concentrated near the bottom (Fig. 1).

Conclusion--The observations discussed under items (2), (3), and (4) show that the 42-mesh sieve size (0.351 mm) divides the grain-sizes in the Enoree River sediment into two groups. The material in these two groups is transported according to different laws. The grain-size of 0.351 mm is also the lower limit of sizes that are found in the bed in appreciable quantities. It is not a mere coincidence that the break in the mechanical-analysis curve of the bed-material happens to occur exactly at the same grain-size that the law of transportation changes; instead one is the cause of the other. The rate of transportation of material coarser than the 42-mesh sieve seems to be some function of the discharge. The reason for this is that if the flow that enters the reach is not "saturated," enough material is available in the bed to saturate the flow without changing the composition and position of the bed sufficiently to influence the rate of transportation. On the other hand, the material finer than a 42-mesh sieve is not found in the bed in sufficient amounts to saturate the flow. The material finer than this size, as obtained by suspended-load sampling, enters the stream from the drainage-basin and merely is being carried past the section.

In the Enoree River all material coarser than the size-limit (42-mesh sieve), whether moving on the bed or in suspension, is moved at a rate of transportation that seems to be a function of the discharge. On the other hand, the material finer than the 42-mesh sieve is merely "washed through" the control-section of the river without deposition. No definite relation exists between the load of this fine material and the stream-discharge, as the quantity of material in the stream depends on drainage-basin factors, such as rainfall, runoff, vegetative cover, tillage-methods, and surface and underground storage.

From the preceding discussion it is evident that while the present generally accepted definitions of "bed-load" and "suspended load" classify the total sediment-load in a convenient and easily understood manner from a qualitative standpoint, it is a classification that is of little value in a quantitative study of the transportation of sediment. Realizing the need for a new basis for the classification of the total sediment-load, special analyses of the Enoree River data were made which resulted in the adoption of the following definitions:

BED-LOAD is that part of the total sediment-load composed of all particles greater than a limiting size (42-mesh sieve for the Enoree River) whether moving on the bed or in suspension, and includes all bed-material in movement.

WASH-LOAD is that part of the total sediment-load composed of all particles finer than the limiting size (42-mesh sieve for the Enoree River) which usually is washed into and through the reach under consideration.

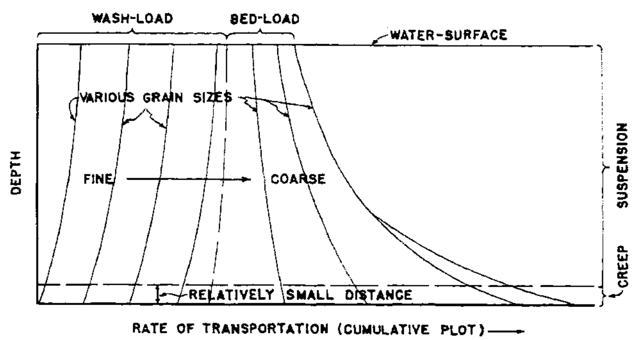

FIG. 4 - ILLUSTRATION OF DISTINCTION BETWEEN THE DIFFERENT PARTS OF THE SOLIDS-LOAD IN A STREAM

Bed-material composition is determined from samples taken in the bottom of the channel in the reach under consideration immediately after a flood has receded to the normal low-water stage. Wash-load may enter a river by sheet-wash, bank-caving, and other causes.

The term "suspension" does not relate to the rate of transportation but merely describes the paths of the grains. In research on the transportation of sand by wind, Bagnold [1] contrasted "flying" grains, which correspond exactly to our suspended load, to grains moving by "surface-creep." He divided the quantity transported into two parts, one part consisting of grains flying at a considerable distance from the ground and the other part consisting of grains creeping or rolling along the ground. This distinction can be used to separate material that can be measured in terms of concentration and material that is measured in terms of rate of accumulation in a suitable trap.

Figure 4 shows graphically the distinction between the different parts of the solids-load as proposed by the above definitions. The total solids-load passing a particular section may be considered to consist of two parts, the bed-load and wash-load. The bed-load consists of material moving as surface-creep and material moving in suspension, both of which can be expressed as a rate related to the stream-discharge. The wash-load, on the other hand, moves almost entirely in suspension and bears no relation to discharge. Primarily the conception is that bed-load and suspended load do not supplement each other, as a certain grain may easily be in suspension and still be considered bed-load.

In many past studies on sediment-load transportation in rivers and flumes having the bed-materials of natural sand-mixtures, including those of very fine-size fractions, difficulty in arriving at definite conclusions occurred because no limit was placed on the size of grains that would be included as bed-load. It is therefore suggested that only those grain-sizes should be included in the bed-load whose movement is a function of the hydraulic characteristics of the stream, or conversely, bed-load functions should be applicable only to those grain-sizes which are found in the bed in considerable quantities.

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THE DIFFERENT APPROACHES TO THE STUDY OF PROPULSION OF GRANULAR MATERIALS AND THE VALUE OF THEIR COORDINATION

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Introduction: Varying patterns of flow--The general property of fluid motion which is most striking to me is its shifting character. If we vary only one factor of a well-defined fluid motion problem, for example, if we vary the opening of the flow from a drinking fountain, we can observe a series of phenomena which differ from each other not only quantitatively, but also qualitatively, that is, in their physical nature and appearance.

Let us examine an example somewhat closer to our problem of propulsion of sand and other granular media by fluids: consider the flow of an infinitely broad turbulent stream around a smooth obstacle. The detailed data given here refer to a cylindrical body of infinite length, this being the shape of the obstacle most completely studied to date. We can distinguish five qualitatively different types of flow, each having a fairly well-defined range of Reynolds' numbers (a, b, c, d, e of Fig. 1). The first type is characterized by steady flow, which is of necessity symmetrical with reference to the flow-direction. With increasing Reynolds numbers, it varies continuously from a flow which is almost symmetrical with respect to the perpendicular to the main flow-direction, until a stage is reached where a strong unsymmetry is caused by two oblong vortices in a fixed position in the rear of the obstacle (Fig. 1a). But vortices lose their stability when they are very oblong, decomposing into several vortices, and thus giving place to a double row of vortices. Von Karman showed the instability of a symmetrical disposition of two parallel vortex-rows, explaining why an "antisymmetric" (alternating) pattern of vortices always is found. These antisymmetric vortex-rows in their turn, for obvious reasons, cannot exist in fixed positions with respect to the obstacle, but must travel downstream. Thus we have here the familiar phenomenon of vortices separating from the body alternately at the sides and giving a perfectly regular pattern of vortex-filaments, each parallel to the axis of the cylinder (Fig. 1b). At a certain distance downstream from the obstacle they gradually lose their identity, giving place to an irregular, turbulent flow. As Fage and Falkner showed experimentally, this turbulent wake resulting from the disintegration of the double row of vortices shows a three-dimensional distribution of vorticity and of velocity-fluctuations in spite of the fact that the double row of vortices itself is, within the accuracy of measurement, a strictly two-dimensional phenomenon.

As the Reynolds number increases and approaches 10,000, the "vitality" of the double row of vortices appears to decrease, the turbulent wake coming nearer and nearer to the obstacles. Above R = approximately 10,000, no row of vortices is noticeable, the turbulent wake starting immediately at the obstacle. A closer investigation shows, however, that this turbulent wake continues to be a product of the separation of a laminar boundary-layer just as the double row of vortices. This laminar boundary-layer on the windward side with the "statistically stable" symmetrical turbulent wake on the lee side persists from R equal about 10,000 until R is of the